Brightloop

Brightloop – Product specializations and Acceptance Criteria

Feature 1: User Registration and Login

1. Purpose

The registration and login system allows users to create accounts and securely access the Brightloop subscription management platform.

2. Functional Requirements

FR1: Users must provide their full name, email address, and password during registration.

FR2: Every registered user must have a unique email address.

FR3: Passwords must contain at least 8 characters.

FR4: During registration, users must be provided with links to the Privacy Policy and Terms & Conditions.

FR5: Users must agree to the Terms & Conditions by selecting a mandatory checkbox before completing registration.

FR6: The Privacy Policy must clearly explain what personal information Brightloop collects, why it is collected, how it is used, and how it is protected.

FR7: The system must prevent registration if the user has not agreed to the Terms & Conditions.

FR8: Users must be able to access the Privacy Policy and Terms & Conditions after registration, such as through the website footer or account settings.

FR9: The system must record the user's acceptance of the Terms & Conditions, including the date and the accepted version.

FR10: Users must be able to log in using their registered email address and password.

FR11: Invalid login credentials must display an appropriate error message.

FR12:Users must agree terms and conditions

FR13: Users must be able to log out securely.

FR14: The system must support different user roles, such as Admin, Staff, and Customer, with appropriate access permissions.

3. Acceptance Criteria

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AC1: A user providing valid registration details can successfully create an account.

AC2: Registration fails when an email address is already registered.

AC3: Passwords shorter than 8 characters are rejected.

AC4: The registration page displays accessible links to the Privacy Policy and Terms & Conditions.

AC5: The Terms & Conditions checkbox is unchecked by default.

AC6: Users cannot complete registration without accepting the Terms & Conditions.

AC7: Clicking the Privacy Policy or Terms & Conditions link opens the correct document.

AC8: Users can access both documents after registration.

AC9: The system correctly records the user's acceptance of the Terms & Conditions.

AC10: Users can log in successfully using valid credentials.

AC11: Incorrect credentials do not allow access to the account.

AC12: After logging out, users cannot access protected pages without logging in again.

AC13: Customers cannot access administrative features.

4. Expected Outcome

Users can securely register, log in, and access only the features permitted for their assigned roles.

Feature 2: Product and Subscription Plan Management

1. Purpose

This feature allows administrators to create and manage products with different variants and subscription plans. Customers can select suitable products and subscribe based on their preferred quantity, price, and billing period.

2. Functional Requirements

FR1: Administrators must be able to create products by entering the product name, description, category, and availability status.

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FR2: Each product must support multiple variants, such as different sizes, quantities, or types.

FR3: Administrators must be able to create multiple subscription plans for each product.

FR4: The system must support four billing periods: daily, weekly, monthly, and yearly.

FR5: Each subscription plan must include a plan name, price, billing period, minimum quantity, and availability dates.

FR6: Administrators must be able to set the minimum quantity required to subscribe to a product.

FR7: Administrators must be able to specify the start and end dates during which a subscription plan is available.

FR8: Customers must be able to view available products, variants, subscription prices, and billing periods before subscribing.

FR9: The system must prevent customers from selecting inactive plans or plans outside their availability dates.

FR10: Administrators must be able to edit product information, variants, prices, and subscription plan details.

FR11: The system must support closing, pausing, and renewing subscriptions according to the configured plan options.

FR12: The system must validate product and plan information and reject invalid entries, such as negative prices, missing product names, or quantities below the required minimum.

FR13: Only authorized administrators must be allowed to create, modify, or deactivate products and subscription plans.

3. Acceptance Criteria

AC1: Administrators can successfully create a product with valid information.

AC2: A single product can have multiple variants.

AC3: Administrators can create daily, weekly, monthly, and yearly subscription plans.

AC4: The system correctly displays the selected product variant, subscription price, and billing period.

AC5: The system prevents subscriptions when the selected quantity is below the minimum requirement.

AC6: Customers cannot subscribe to plans outside their specified availability dates.

AC7: Administrators can successfully update product and plan information.

AC8: The system supports closing, pausing, and renewing subscriptions according to the available options.

AC9: The system rejects invalid prices, missing required information, and invalid quantities.

AC10: Customers cannot access administrative product and plan management features.

4. Expected Outcome

Administrators can efficiently manage products, variants, and subscription plans with flexible billing periods and availability settings. Customers can view available products and select subscriptions according to their preferences, while the system ensures accurate validation and access control.

Feature 3: Quotation to Subscription

1. Purpose

This feature allows businesses to create quotations for customers and convert accepted quotations into subscriptions. It ensures that subscription details, pricing, validity periods, and statuses are properly managed.

2. Functional Requirements

FR1: Authorized users must be able to create quotations for customers by selecting products, variants, subscription plans, and quantities.

FR2: Each quotation must include customer details, selected products, quantities, prices, billing periods, and total amount.

FR3: Administrators or authorized staff must be able to create quotation templates for frequently offered subscription packages.

FR4: Each quotation must have a validity period specifying the number of days for which the offer remains valid.

FR5: The system must prevent expired quotations from being confirmed without renewal or an updated quotation.

FR6: Every subscription must have a unique subscription number for identification and tracking.

FR7: A subscription must support multiple line items, allowing different products or variants to be included.

FR8: The system must support the following subscription statuses:

- Draft
- Quotation
- Confirmed
- Active
- Closed

FR9: Authorized users must be able to convert an accepted quotation into a confirmed subscription.

FR10: The system must update subscription statuses according to the subscription lifecycle.

FR11: Customers must be able to view their own quotation and subscription details.

FR12: The system must prevent unauthorized users from modifying or confirming subscriptions.

3. Acceptance Criteria

AC1: Authorized users can successfully create quotations with valid customer and product information.

AC2: The system correctly displays quotation details, including prices, quantities, and billing periods.

AC3: Quotation templates can be created and reused.

AC4: Each quotation has a clearly defined validity period.

AC5: Expired quotations cannot be confirmed without being renewed or updated.

AC6: Every subscription receives a unique subscription number.

AC7: Multiple products or variants can be included in a single subscription.

AC8: The system correctly displays Draft, Quotation, Confirmed, Active, and Closed statuses.

AC9: Accepted quotations can be converted into confirmed subscriptions.

AC10: Customers can view only their own quotations and subscriptions.

AC11: Unauthorized users cannot modify subscription details.

4. Expected Outcome

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The system provides an organized process for creating quotations, managing their validity, and converting accepted quotations into active subscriptions. Each subscription can be tracked using a unique number and clearly defined statuses.

Feature 4: Invoice Generation and Billing Management

1. Purpose

This feature enables businesses to automatically generate and manage invoices for customer subscriptions. It ensures accurate billing, tax calculations, payment tracking, and timely invoice generation according to the selected subscription plan.

2. Functional Requirements

FR1: The system must automatically generate invoices based on active customer subscriptions and their billing periods.

FR2: Each invoice must contain a unique invoice number, customer details, subscription number, product details, quantity, price, invoice date, and due date.

FR3: The system must support daily, weekly, monthly, and yearly billing cycles.

FR4: Applicable taxes must be calculated automatically based on the configured tax rules.

FR5: The system must calculate the total invoice amount, including applicable taxes and discounts.

FR6: The system must support the following invoice statuses:

- Draft
- Confirmed
- Paid

FR7: Authorized users must be able to review and confirm draft invoices before finalization.

FR8: The system must allow authorized users to send invoices to customers.

FR9: Invoices must be available in a printable format.

FR10: The system must prevent duplicate invoice generation for the same subscription and billing period.

FR11: The system must ensure that scheduled billing periods are not skipped.

FR12: Customers must be able to view their own invoices and payment status.

FR13: The system must prevent unauthorized users from modifying invoice information.

FR14: The system must correctly identify paid and outstanding amounts based on recorded payments.

3. Acceptance Criteria

AC1: An invoice is automatically generated when an active subscription reaches its scheduled billing date.

AC2: Each invoice contains the correct customer, product, subscription, and billing information.

AC3: The system generates invoices according to the selected daily, weekly, monthly, or yearly billing cycle.

AC4: Applicable taxes are calculated correctly.

AC5: Discounts and taxes are reflected accurately in the final invoice amount.

AC6: Invoices correctly display Draft, Confirmed, and Paid statuses.

AC7: Authorized users can confirm and send invoices.

AC8: Invoices can be printed with their details clearly displayed.

AC9: The system does not generate duplicate invoices for the same subscription and billing period.

AC10: Scheduled invoices are not skipped, including when a billing process is delayed or restarted.

AC11: Customers can view only their own invoices.

AC12: The system correctly displays paid and outstanding amounts based on recorded payments.

4. Expected Outcome

The system provides accurate and automated subscription billing by generating invoices on schedule, applying relevant taxes and discounts, maintaining invoice statuses, and preventing duplicate or missed billing. Businesses can efficiently manage invoices while customers can access clear and reliable billing information.

Feature 5: Discount Management

1. Purpose

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This feature allows administrators to create and manage discount rules for subscription plans. It ensures that eligible customers receive the correct discounts based on predefined conditions, including discount type, minimum purchase requirements, validity dates, and usage limits.

2. Functional Requirements

FR1: Only administrators must be allowed to create, edit, activate, or deactivate discount rules.

FR2: The system must support two types of discounts:

- Fixed amount discounts
- Percentage-based discounts

FR3: Each discount rule must contain a name, discount type, discount value, validity start date, and expiry date.

FR4: Administrators must be able to specify minimum purchase amounts or quantities required for a discount.

FR5: Administrators must be able to set usage limits for each discount rule.

FR6: The system must automatically apply eligible discounts to the customer's next invoice.

FR7: Expired or inactive discounts must not be applied.

FR8: Discounts that have reached their usage limits must not be applied.

FR9: The system must verify all discount conditions before applying a discount.

FR10: The invoice must clearly display the original amount, discount applied, and final amount payable.

FR11: The system must prevent discounts from reducing the invoice subtotal below zero.

FR12: Only authorized users must be able to modify discount rules.

3. Acceptance Criteria

AC1: Administrators can successfully create fixed amount and percentage-based discount rules.

AC2: The system correctly calculates discounts according to the selected discount type.

AC3: Discounts are applied only when minimum purchase conditions are satisfied.

AC4: Discounts are applied only within their validity dates.

AC5: Discounts cannot be applied after reaching their usage limits.

AC6: Eligible discounts are automatically reflected in the next invoice.

AC7: The invoice correctly displays the original amount, discount amount, and final payable amount.

AC8: Invalid discount values, including negative amounts and percentages above 100%, are rejected.

AC9: Customers and unauthorized staff cannot create or modify discount rules.

AC10: Applying a discount never produces a negative invoice subtotal.

4. Expected Outcome

The system provides a reliable and automated discount management process that allows administrators to configure promotional offers while ensuring accurate calculations, eligibility verification, and proper application to customer invoices.

- Subshare – Sample Data

1. Product Details

The following fictional products are provided for testing product creation, variant selection, and subscription management.

Product ID	product Name	Category	Available variants
P001	Fresh milk	Diary	500 ml,1 litre
P002	Coffee Powder	Beverages	250g,500g
P003	Newspaper	Publications	English,Malayalam
P004	Mealbox	Food Services	Regular,Premium
P005	Cloud storage	Digital Services	50GB,100GB

All five products are initially marked as active for testing purposes.

2. Subscription Plan Details

The following six fictional subscription plans are prepared for testing Brightloop's product and subscription management features.

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Plan ID| Product| Variant| Plan Name| Billing Period| Price (₹)| Minimum Quantity
PL001| Fresh Milk| 500 ml| Daily Milk Plan| Daily| 30| 1
PL002| Fresh Milk| 1 litre| Monthly Milk Plan| Monthly| 1,800| 1
PL003| Coffee Powder| 250 g| Weekly Coffee Plan| Weekly| 250| 1
PL004| Newspaper| English| Monthly News Plan| Monthly| 200| 1
PL005| Meal Box| Regular| Daily Meal Plan| Daily| 120| 2
PL006| Cloud Storage| 100 GB| Annual Cloud Plan| Yearly| 1,999| 1

Plan Availability

- Start Date: 1 November 2026
- End Date: 31 December 2027
- Initial Status: Active
- Pause Option: Enabled
- Renewal Option: Enabled
- Close Option: Enabled

These settings are proposed sample values and must be confirmed with the development team.

Expected System Behaviour

1. Administrators can create and edit these subscription plans.
2. Customers can view active plans and their prices.
3. The system displays the correct billing period.
4. Customers cannot subscribe below the specified minimum quantity.
5. Plans outside their availability dates cannot be selected for new subscriptions.
6. Subscriptions can be paused, renewed, or closed according to the configured options.